

Make the Qubit Tune Itself

Closed-loop auto-calibration of a drifting three-level transmon, and what it does to a QAOA

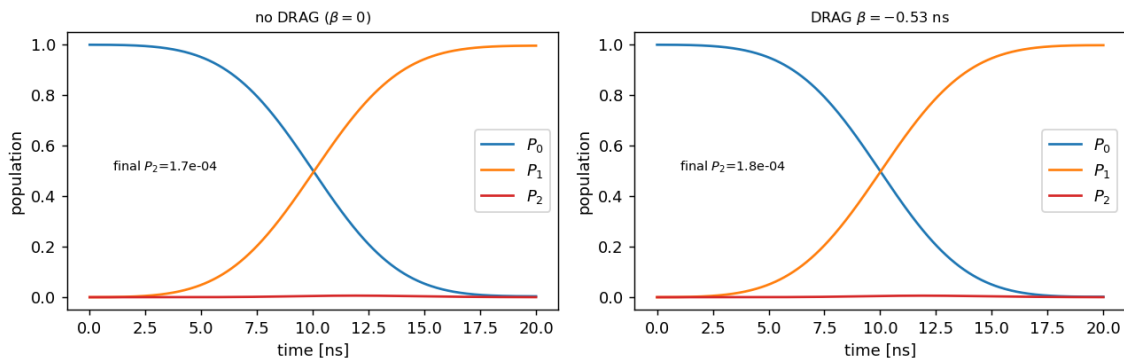
What we built

- Three-level transmon with drifting detuning (OU + telegraph) and drive gain (sinusoid + OU), seed 2026
- Agent sees only shot counts (500 μ s per shot, readout errors 1% / 3%)
- Primitives: amplitude (error amplification), Ramsey frequency (Bayesian), DRAG
- Policies: P0 never, P1 fixed period, P2 health check then recalibrate
- Judged by the hidden oracle gate error and a p=1 QAOA on a 4-ring

drift $\rightarrow$ gate $\rightarrow$ counts $\rightarrow$ estimate $\rightarrow$ decision $\rightarrow$ new controls (pulse only)

Plant check: the static π pulse

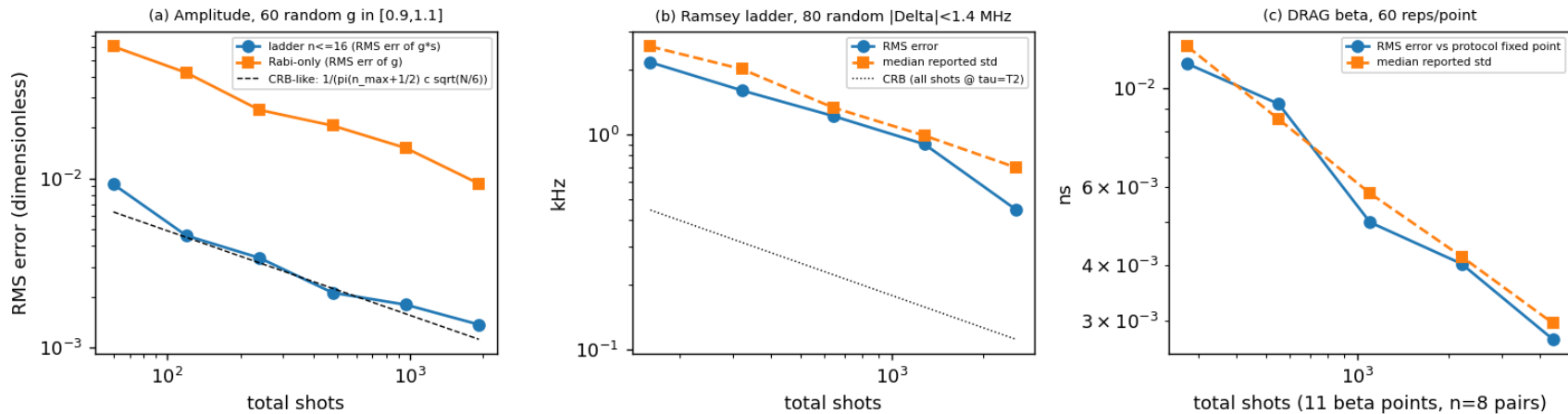
Fig. 3 Static Gaussian pi pulse ($\Delta=0$, $g=1$): populations during the gate, starting in $|0\rangle$. Exact propagation, 400 steps.



- $\beta = 0$: error $2.5e-3$, leakage $1.7e-4$
- Best $\beta = -0.313$ ns (first order: -0.53 ns)
- Best error: $1.75e-4$ (π), $1.4e-5$ ($\pi/2$)
- Propagator: unitary to $1e-13$, 2nd-order convergent

Calibration primitives converge

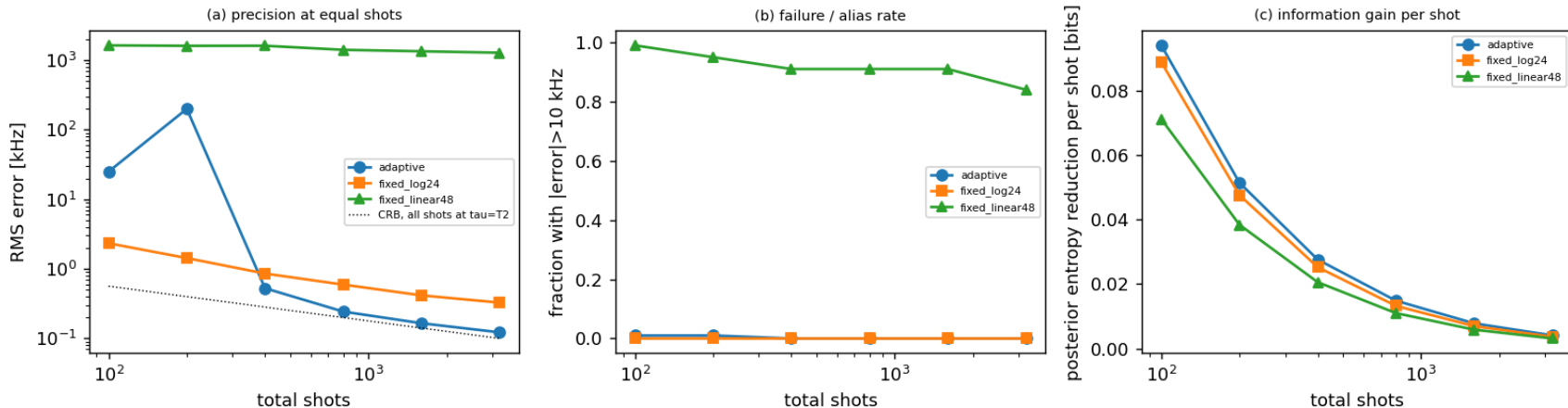
Fig. 4 Calibration-primitive convergence, static plant. Points are RMS over repetitions (reps in titles); no error bars.



- Amplitude: RMS $9e-3 \rightarrow 1.4e-3$ ($60 \rightarrow 1920$ shots); Ramsey ladder aliases beyond 1.67 MHz, as predicted
- DRAG: $0.011 \rightarrow 0.0027$ ns; order matters: frequency first

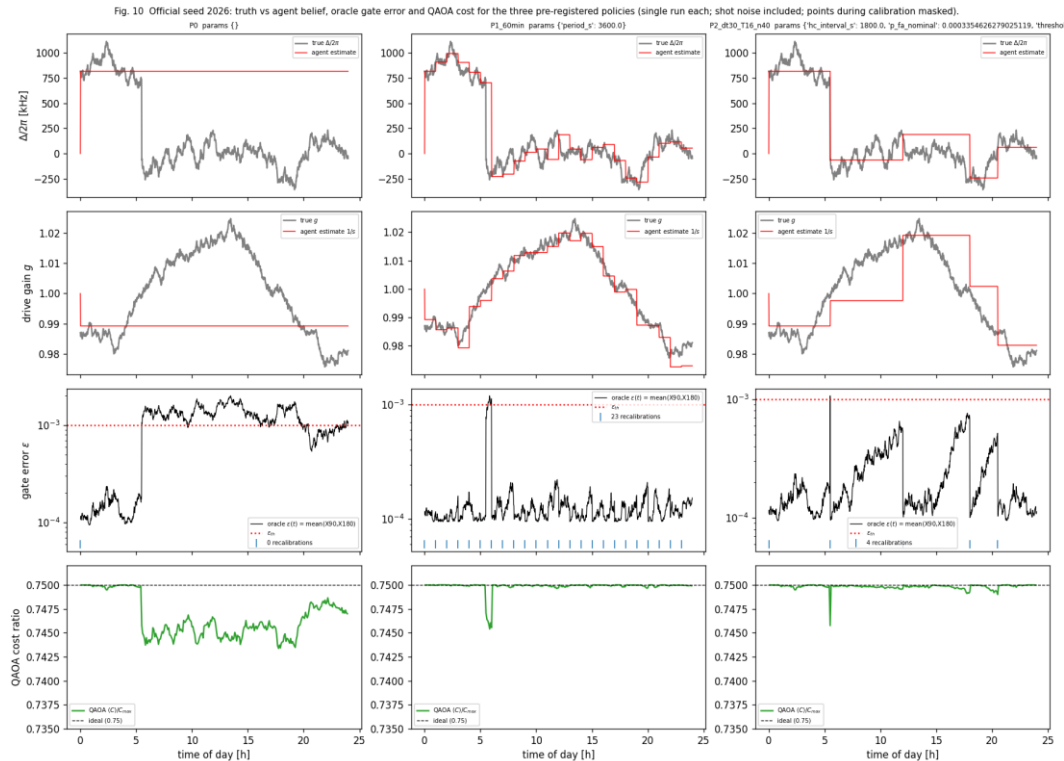
Adaptive Bayesian Ramsey vs fixed grid

Fig. 8 Adaptive Bayesian vs fixed delay grids, equal total shots. 100 random true detunings $U(\pm 1.2 \text{ MHz})$ per budget; RMS is not a CI.



- 0.5 kHz RMS: ~420 shots adaptive vs ~1,110 fixed log grid (2.7 $\times$)
- Honest limit: at $N \leq 200$ adaptive has ~1% alias failures; fixed grid is safer
- Uniform linear grid aliases (84–99% failures)

One day, official seed 2026

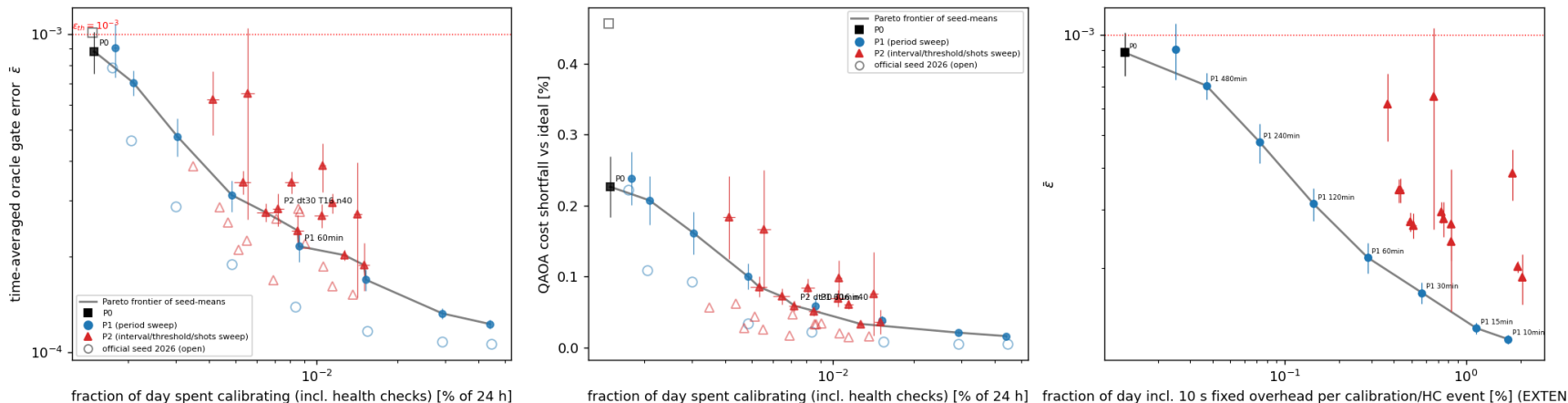


- P0: error above $1e-3$ for much of the day after the jump
- P1 (60 min): recovers within the hour
- P2: detects the jump, then recalibrates 3 more times
- Cold calibration: $\sim 2,600$ shots (1.3 s); warm: ~ 520 shots (0.26 s)

Truth (grey) vs agent belief (red). Single run per policy, shot noise included.

Pareto: cost vs gate error (21 seeds)

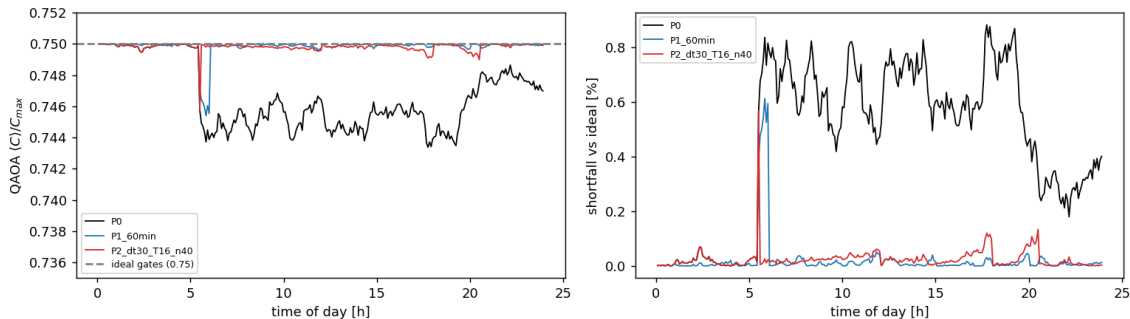
Fig. 11 Pareto analysis. 21 random drift seeds (1..21): points = across-seed mean, bars = 95% Student-t CI of the mean (both axes). Open markers: official seed 2026. (c) is a sensitivity extension not in the spec.



- P1 60 min: error $2.15e-4 \pm 0.23e-4$ (95% CI) vs $8.84e-4$ for P0, at 0.0086% of the day
- P2 gives no clear gain when health-check shots count as downtime: $+6.7e-5$ error for 0.0014 points less time
- With a 10 s overhead per event P2 leaves the frontier; if checks are free it fills the middle

From hardware error to QAOA cost

Fig. 12 $p=1$ QAOA (4-ring MaxCut, $\gamma=\pi/4$, $\beta=\pi/8$, ideal RZZ) with instantaneous 2×2 blocks $M(t)$; official seed 2026, 5-min sampling.



- Ideal $\langle C \rangle / C_{\max} = 0.75$
- Policy spends y of the day calibrating; QAOA within $z = 0.5\%$ of ideal for w of the day
- Telegraph jump hurts QAOA most (0.176% shortfall)

Policy	y	w ($z=0.5\%$)
P0	0.0015%	88.0%
P1, 60 min	0.0086%	97.2%
P2, 30 min	0.0072%	97.1%

Takeaways and limits

- Recalibrating every 30–60 min keeps error near the $0.94e-4$ floor at under 0.02% of the day
- Frequency jump hurts the algorithm most; gain drift hurts raw gate error most
- Adaptive Bayesian design saves $\sim 2.7\times$ shots at ≥ 400 shots; not at tiny budgets
- Negative result: health-check policy gives no clear gain over periodic recalibration under the specified accounting

Limits

- Oracle is coherent only; idle-detuning phase in QAOA not modelled
- Sweep ranges chosen after looking at the official seed; only 3 headline policies pre-registered
- Developed with AI assistance (Claude, Anthropic); cited in the report